GROUP_03 · W1 Report

Discovering Questions Worth Investigating — Business Intelligence (IIB423T-1)
Universidad del Desarrollo · Facultad de Ingeniería
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We are working with four of the course's public MINSAL/FONASA health datasets (Urgencias, REM, Establecimientos, GRD; 2020–2024). We left out cartografía_censo_2024 for now since neither candidate below needs population normalisation yet. Below are our two candidate lines, a quick look at real data, our comparison, and where we stand so far.

1. Candidate A — Emergency care by age group and facility type

Problem: how emergency care demand (2020–2024) splits across age groups (under 1, 1–4, 5–14, 15–64, 65+) and facility types (Hospital, SAR, SAPU, SUR, CESFAM), and whether some age groups end up mostly in lower-resolution facilities instead of hospitals.

Why it matters: MINSAL's Estrategia Nacional de Salud 2021–2030 (Res. Ex. N°831, June 2024) lists "reducing health inequities" as one of its four Sanitary Objectives. If older adults or infants end up mostly in SAPU/SUR instead of hospital ER, that looks like an age-access gap tied to that objective. (Ministerio de Salud de Chile, minsal.cl/estrategia-nacional-de-salud-2021-2030/)

Question / possible use: which age groups get a bigger share of low-complexity care instead of hospital ER, and does this change by region? A Health Service network planner could use this to decide where SAPU/SAR capacity needs reinforcing.

Data check so far: we loaded the real 2024 Urgencias file (*AtencionesUrgencia2024.csv*) — 8,973,229 rows, 21 columns. It confirms the fields we need are actually there: **Total**, the five age columns (Menores_1 … De_65_y_mas), **GLOSATIPOESTABLECIMIENTO**, **GlosaCausa**, **NombreRegion**, **NombreComuna** and **semana**. One quick thing we noticed: some rows are sub-totals (e.g. "TOTAL DEMÁS CAUSAS", "SECCIÓN 2. TOTAL DE HOSPITALIZACIONES"), so any cause-level sum needs to filter those out first or numbers will double-count. We still need to run the actual age-by-facility-type breakdown for C1 — ran out of time this week to finish that part.

Limitation: no triage/severity field in this dataset, so we can only look at volume, not clinical urgency.

Later on (ML/dashboard): could try classifying communes by "age-access risk" later on, and a simple dashboard for a regional planner showing age share by facility type over time.

2. Candidate B — Public vs. private gap in hospital case complexity

Problem: do public facilities end up with more complex/severe hospital cases (GRD) than private ones, relative to how much they attend?

Why it matters: a Salud Problema (UAM Xochimilco) article argues that despite MINSAL's equity strategies, AUGE/GES has pushed resources toward the private sector, reinforcing a public/private divide. Worth checking if that shows up in the actual complexity data.

Question / possible use: do public hospitals carry a bigger share of high-weight/high-severity GRD cases than private ones? Could inform a FONASA analyst looking at case-load balance.

Data check so far: we inspected one real GRD row (course file SA_21_V1.2). One row = one hospital discharge, with **PREVISION** (e.g. FONASA / Isapre) and grouper fields **IR_29301_PESO** (relative case weight) and **IR_29301_SEVERIDAD** right there — no join needed for a first cut. In our sample row, PESO was 0.3045 for a routine ambulatory surgery case, which seems plausible but we haven't checked it against a full file yet. We have not loaded the full GRD file with real numbers yet, this is the next step for C1.

Limitation: PESO is a relative weight, not a cost in pesos, so we can talk about complexity but not about money/spending.

Later on (ML/dashboard): maybe predict which public hospitals are heading toward heavier case-loads; dashboard for FONASA showing case-mix weight split by PREVISION.

3. Comparing the two

	A – Age / facility type	B – Public vs. private
Support	MINSAL Estrategia Nac. Salud 2021-2030	Salud Problema article (UAM)
Data fit	Strong – fields confirmed in real 8.9M-row file	Strong on paper – confirmed in one real row, full file still pending

	A – Age / facility type	B – Public vs. private
Main gap	No triage/severity field	PESO is relative, not cost; full file not loaded
Next step for C1	Age x facility-type breakdown, by region	Load full GRD file, group by PREVISION

Where we lean right now: Candidate A, mostly because we already have the full real dataset loaded and confirmed workable at scale (8.9M rows), while B is still resting on a single sample row. Not a final call — open to switching if the GRD file turns out easier to work with once we open it fully.

Questions for the instructor:

- Is a relative GRD weight (PESO) an acceptable stand-in for complexity, given we have no cost figure?
- For Candidate A, do you expect us to normalise by commune population from the start, or is a raw-share view okay for C1?

Process Record

Decisions and checks: Dropped cartografía_censo_2024 (not needed yet). Narrowed 10 brainstormed lines to these 2 based on which fields we could actually confirm in real files. Loaded the full 2024 Urgencias CSV and confirmed all needed columns exist. Inspected one real GRD row to confirm PREVISION/PESO/SEVERIDAD exist. Both join keys (idestablecimiento, COD_HOSPITAL) are proposed only, not tested yet.

Instructor feedback: none received yet as of this submission.

Contributions: Miguel Figueroa – Candidate A (sources, Urgencias data check). Sebastián Bunzli – Candidate B (sources, GRD data check), report assembly.

AI use: Used Claude (Anthropic) to fix local SSL/PowerShell errors when downloading the course data, to help read the data dictionaries and one real GRD row, and to help draft/organise this report. We checked every source cited here ourselves before using it. We wrote this report quickly under time pressure and focused effort on making sure the field names and data checks are real, not on polishing the writing.

References

- Ministerio de Salud de Chile. Estrategia Nacional de Salud para los Objetivos Sanitarios al 2030 (Res. Ex. N°831, 2024). minsal.cl/estrategia-nacional-de-salud-2021-2030/
- Salud Problema (UAM Xochimilco). Public–private duality in the Chilean health system. saludproblemaojs.xoc.uam.mx
- DEIS/MINSAL. Diccionario de Datos, Atenciones de Urgencia (SADU) and Establecimientos 2021. deis.minsal.cl
- FONASA. Base GRD (SA_21_V1.2), course-provided extract.